Example – 3

The difference in head between two ends of a pipeline, 400 m long and 200 mm diameter is 2 m. Find out the discharge through the pipe by taking friction factor as 0.003. Neglect minor losses.

Given data:

Length of pipe (l) = 400 m

Diameter of pipe (d) = 200 mm

= 0.20 m

Difference in head (hf) = 2 m

Friction factor (f) = 0.003

To find out:

Discharge through the pipe, Q

Solution:

Loss of head due to friction,

$$h_f = \frac{flQ^2}{12d^5}$$

$$\text{i.e., } 2 = \frac{0.003 \times 400 \times Q^2}{12 \times (0.20)^5}$$

$$2 \times 12 \times (0.20)^5 = 0.003 \times 400 \times Q^2$$

$$\therefore Q^2 = \frac{2 \times 12 \times (0.20)^5}{0.003 \times 400} = 0.0064$$

$$Q = \sqrt{0.0064} = 0.08 \ {}^{m^3}/_{\text{sec}}$$

Result:

Discharge through the pipe

$$(Q) = 0.08 \ {}^{m^3}/_{\text{sec}}$$

7.5.8 Chezy's formula for velocity of flow in pipes

Velocity, $v = c\sqrt{mi}$

Where,

v = velocity of flow

c = chezy's constant

m = Hydraulic mean depth

i = Hydraulic gradient

Hydraulic mean depth

$$(m) = \frac{Area\ of\ pipe (A)}{wetted\ perimeter (p)}$$

Wetted perimeter of pipe (p) = πd

$$\text{Area of pipe (A)} = \frac{\pi d^2}{4}$$

$$\therefore m = \frac{A}{p} = \frac{\left(\frac{\pi d^2}{4}\right)}{\pi d} = \frac{d}{4}$$

Hydraulic mean depth, $m = \frac{d}{4}$

Hydraulic gradient, $i = \frac{h_f}{l}$

7.5.9 Chery's formula for loss of head in terms of velocity

We know, Velocity, 'v' = $c\sqrt{mi}$

$$v = c\sqrt{m \times \frac{h_f}{l}} \qquad \left(i = \frac{h_f}{l}\right)$$

Squaring on both sides,

$$v^2 = c^2 \times m \times \frac{h_f}{l}$$

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